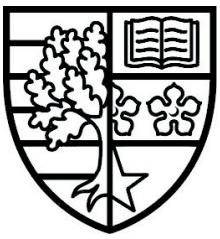
Self-Supervised Learning for Analysis of Facial Micro-expressions

by Chong Kian Fei

Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

Supervised by Prof. John See



Heriot-Watt University
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Abstract

Facial Micro-Expressions (FMEs) are brief and involuntary facial movements that reveal genuine emotions. However, these FMEs are difficult for the human eye to perceive and require expert knowledge to recognise them. Existing automated micro-expression recognition models have attempted to recognise FMEs but have been hindered by the scarcity of large-scale, labelled datasets. To address this challenge, this study proposes a novel Self-Supervised Learning (SSL) framework for analysing FMEs. It learns robust spatio-temporal features from unlabelled video, which eliminates the need for costly manual annotations. This study employs a two-stage pre-training approach on CAS(ME)^3 that combines Generative Learning (GL) for learning facial motion dynamics and Contrastive Learning (CL) for discriminating emotion pairs for each category. The model will be fine-tuned on the CASME II dataset to classify emotions into three standard categories (Positive, Negative, or Surprise). A feasibility study was conducted to validate the GL stage for this study. The feasibility study demonstrated that the model can accurately reconstruct the middle frame, learning robust feature learning without the need for costly manual annotations of apex frames. This study establishes a method to overcome the data scarcity in MER by utilising a large-scale unlabelled dataset without the need for an apex frame.

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1 Introduction

Human communication for expressing thoughts usually involves not only spoken language, but also body language. Among these body languages, facial expression is a primary channel for conveying people's emotional states and human intentions [49]. People often consciously control their facial expressions to gain social advantages, hiding their true feelings, which gives persuasiveness to others. However, involuntary responses still reveal genuine emotions through subtle and rapid facial muscle movements. These rapid involuntary signals, also known as facial micro-expressions, last for less than 500 milliseconds and are difficult for the human eye to perceive [49]. Furthermore, facial micro-expressions have gained attention in psychology-related fields, security, and even normal human conversation due to its potential in understanding concealed human thoughts [32, 46, 49].

1.1 Background and Motivation

People with the ability to read these facial micro-expressions can bring a huge improvement to their quality of life [14], but the difficulty for humans to perceive those subtle and fleeting facial micro-expressions is extremely high. People who want to develop the ability to spot these movements find it exceptionally time-consuming and requires expert knowledge. Indeed, some research has attempted to train individuals on learning FMEs in a controlled psychology lab, but it has also encountered difficulties and proved time-consuming [8]. To address these challenges, existing research has developed an automated system to assist in learning facial micro-expressions. However, those supervised learning approaches rely heavily on the manual and precise identification of the apex frame, which is the peak moment of micro-expressions intensity [1, 5, 49]. This annotation process is not only time-consuming but also costly, increasing the cost of labelling a dataset.

To address these challenges, this study proposes a self-supervised learning (SSL) model that is capable of learning spatio-temporal features from unlabelled videos without the use of apex frames. This model aims to achieve a high generalisation capability across different datasets without apex frame labelling, eliminating the need for costly apex frames.

1.2 Aim and Objectives

Aim: Develop a self-supervised learning model that are capable for spotting and recognising facial micro-expressions without the need of apex frame annotations from unlabelled video. The model should exhibit high generalisation capability across different datasets and be able to be benchmarked by its performance against existing models to demonstrate effectiveness.
Objectives:

1. Implement an architecture backbone as a feature extractor for the model.
2. Pre-train the model with unlabelled video as an upstream task to capture robust features and form pseudo labels for each emotion.
3. Fine-tune the model with labelled video as downstream task to classify emotion-pair
4. Evaluate and optimise the model to increase effectiveness.

1.3 Organisation

The following illustrates the report's structure:

1. **Introduction:** Gives an overview of the report and explains the background, motivation, research aim and objectives of this study.
2. **Background Research:** Examines existing studies and related work on facial micro-expression recognition research.
3. **Methodology and Work Plan:** Describes the framework of this study, such as backbone architecture selection, pre-training and fine-tuning strategy, performance matrix analysis method, and dataset characteristics.
4. **Feasibility:** Discusses the feasibility of this study by examining the tools being used for model implementation, dataset selection and the summary of early testing.
5. **Conclusions:** Summarises the findings of the study and outlines its future directions.

2 Background Research

This section provides a review of related work on micro-expressions recognition (MER), tracing the evolution from handcrafted features to deep learning and self-supervised learning techniques. It highlights how these techniques extract spatial and temporal features captured from the facial and discusses their limitations, which motivated the research community's shift toward deep learning and self-supervised learning techniques.

2.1 Micro-expression Recognition

Local Binary Patterns from Three Orthogonal Planes (LBP-TOP) [50] is one of the traditional methods used to study facial micro-expressions. LBP-TOP captures local spatial feature (appearance) and temporal feature (motion) by extracting the pattern across three orthogonal planes in the video [50]. These features are used to estimate emotional states based on subtle facial movements over time.

Zhao and Pietikainen [50] demonstrates the success of the LBP-based features in recognising facial expressions in the research field. Additionally, Support Vector Machines (SVM) [4], a statistical technique commonly used for binary classification tasks, often combined with LBP-based feature extraction method. The LBP-TOP combined with SVM has also been widely used for recognising facial micro-expression and classifying emotion pairs.

A novel approach introduced a method that utilises a 200fps high-speed camera for capturing facial motion and creating a model to recognise facial micro-expressions [35]. It separates the face into 12 regions to categorize the micro-expressions. This approach inspired the concept that local regions are key to success in MER.

Traditional feature-based methods, such as LBP-TOP combined with SVM [1], offer excellent performance in capturing local spatial (appearance) and temporal (motion) features. The key advantage of these hand-crafted feature methods was their computational efficiency. For example, LBP-TOP was designed to be computationally simple [50] but the

limitation of these methods is their over-dependency on hand-crafted feature extractors [40]; these features are static for one dataset, which leads to a restricted generalisation capability for other datasets in real-world scenarios. To overcome this challenge, the research community has shifted towards Deep learning approaches, which automatically learn a hierarchy of features directly from video.

2.2 Deep learning approach

Facial micro-expression recognition (MER) has been significantly achieved through deep learning architectures, which provides the advantage of processing and integrating both spatial and temporal information from video sequences layer by layer.

Traditional Convolutional Neural Network (CNN) models have been established as a powerful class of models for image recognition problems, demonstrating the understanding of image content from a single frame [17], and showing that they are capable of learning powerful features from video frames [17]. For example, a ResNet-like model [20], which is a CNN-based architecture, has demonstrated great performance.

The strength of the CNN architecture makes it adapt to capture spatio-temporal features or motion feature extraction [25]. CNNs can be extended to 3D CNNs, such as Pseudo-3D Residual Network [36], which are a powerful model for micro-expression recognition. Using CNNs as feature extractors is also a great approach for a micro-expressions recognition model, making it a critical and effective in MER [25, 36].

Another model is a Vision Transformer-based model, a revolutionary deep learning architecture that was originally designed for Natural Language Processing tasks [3]. It uses Self-attention, which allows each part of the image to compute its relationship with other parts of the image regardless of distance. Unfortunately, ViT is only performed effectively when large training datasets are available [3]. Given the limitation of this study's scope, training a ViT is impractical since it may lead to a high risk of poor performance or overfitting of the model.

Furthermore, the Generative Adversarial Network (GAN) has demonstrated remarkable effectiveness as a deep learning approach in recent research [22, 26, 53]. It trains both generator and discriminator through adversarial learning to achieve data distribution estimation and new sample creation [26]. The approach proposed in [53] employs an unsupervised generator to create the fake images, while the discriminator enhances its ability to distinguish between images generated by the generator and real images [53]. However, even though these approaches achieved excellent performance, they are also widely known to be unstable to train without a simplified attention mechanism [22] and highly sensitive to hyper-parameter settings, which have constrained the dataset's capability [53]. This sensitivity contradicts this study's goal of creating a model with

high generalisation capability for different datasets. Therefore, a self-supervised learning approach is adopted and detailed in the next sections.

2.3 Self-supervised Learning

The primary limitation of deep learning is its heavy dependency on labelled data [40], which requires costly manual annotation. To address the challenges of expensive and time-consuming human-labelled datasets, Self-Supervised Learning (SSL) has been introduced to overcome the challenge of data scarcity [19, 52]. SSL provides a way of representation learning without requiring annotations [38] by forcing the model to learn features from unlabelled data and generating pseudo labels [52]. Based on the approach by Schiappa et al. [38], SSL can be categorised into four types of approaches: contrastive learning, generative learning, pre-text learning, and multimodal learning. Each exhibits distinct characteristics and capabilities across different tasks.

Contrastive Learning
Contrastive learning (CL) is a SSL technique that trains a model to differentiate between similar and dissimilar samples by pulling similar (positive) pairs closer together and pushing dissimilar (negative) pairs farther apart [38]. CL is a great classification method and feature learning approach for micro-expression recognition, and has been demonstrated as an effective framework in learning useful representations from unlabelled data [19]. Recent approaches like HDRCL [54] and CMNet [43], using contrastive learning framework has demonstrated remarkable performance on micro-expression recognition.

Generative Learning
Another SSL technique is Generative Learning (GL), which utilises the generation capabilities of a neural network to synthesise the video [38]. GL usually masks a part of the input, and the model is forced to reconstruct the masked part to learn the robust representation of the data. Some of the successful models are the GANs [26] mentioned in Section 2.2 and Masked Autoencoders (MAE) [47]. While GANs use a generator and a discriminator to train the model, MAE employs an encoder-decoder model to learn general representations from reconstruction. Zhang et al. [47] demonstrated another method named BEiT also outperforming, using a mask-then-predict strategy to train the model.

Pre-text Learning
Pre-text Learning (PL) is another SSL technique that provides the neural network with predefined tasks, such as frame prediction, temporal order prediction, playback speed prediction, and video jigsaw [38]. The model acquires features by solving these preliminary, or pre-text tasks and subsequently applies knowledge obtained to downstream tasks. For example, Fernando et al. [12] demonstrated an Odd-One-Out network that assigns the model a validation task of identifying the anomalous video frame within a temporal sequence. Another method proposed by Jing et al. [16] involves predicting the rotated frame

to facilitate learning of video motion.

Multimodal Learning
Multimodal Learning (ML) is a framework for learning features by comparing different modalities, such as video and text or video and audio [38]. The model will be given a validation task to determine whether a video clip and an audio snippet are synchronised from the same source, categorising them as a positive pair if they are and vice versa [55]. An audio-video approach has demonstrated the effectiveness of multimodal learning by using pseudo-labels generated from one modality to guide learning in the other modality [2]. Additionally, video-text multimodal learning has proven effective for learning video representations by fusing the expert features from multiple modalities [13].

2.4 Discussion

Traditional methods such as LBP-TOP, which rely on hand-crafted features, have demonstrated effectiveness in capturing spatio-temporal characteristics. However, their limitation in automatically learn features from video limits their generalisability and has prompted a shift towards deep learning approaches. While deep learning methods like GANs or ViT achieve superior performance, but they require large annotated datasets. To mitigate the high cost of annotation, self-supervised learning enables training feature-extraction backbone networks using unannotated data, eliminating the need for manual labelling.

3 Methodology and Work Plan

This section outlines the overall methodology and development plan for the proposed Self-Supervised Learning for Analysis of Facial Micro-expressions model. It describes the overall framework and timeline, model architecture, and dataset comparison and selection.

3.1 Timeline of the study

The whole study will undergo two semesters. The detailed work plan is presented in Figure 10 in Section A.1 and Figure 11 in Section A.2. During the first semester, the project scope and objectives are finalised, a comprehensive literature review for the topics is completed, and planning for the whole study is established. The second semester will mainly focus on the implementation of the model for this study, followed by evaluation and refinement to increase performance. At the end of Semester 2, a complete documentation of the study with analysis will be done, providing the most accurate information.

3.2 Overall Framework

The overall framework for developing the model for ”Self-Supervised Learning of Analysis for Facial Micro-expression“ is divided into two stages, which are pre-training and fine-tuning, supported by several evaluation and optimisation phases.

As illustrated at 1, the framework of this methodology can divided into five-step:

1. **Backbone Architecture Selection:** Selecting a suitable neural network for model.

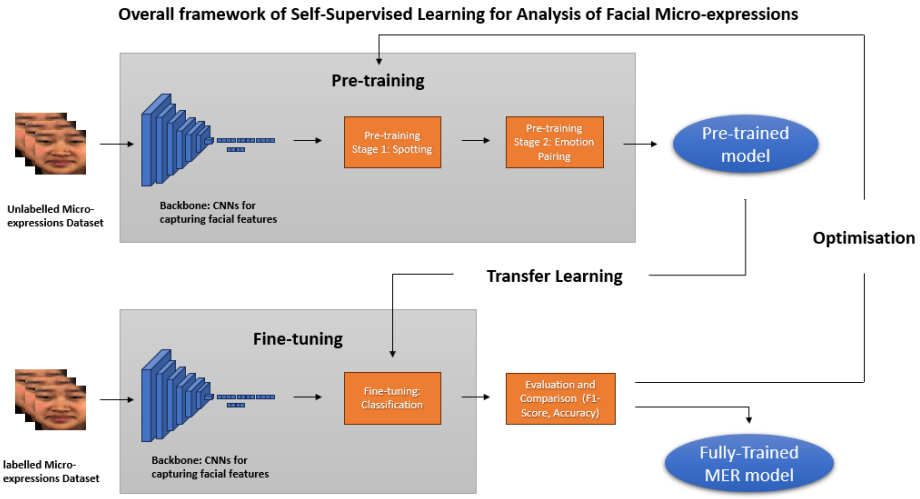


Fig. 1. Overall Framework of the Self-Supervised Micro-Expression Recognition Model

2. **Pre-training Stage 1 Planning:** Training the model on unlabelled data for spotting micro-expressions.
3. **Pre-training Stage 2 Planning:** Training the model on unlabelled data for discriminative the emotion pair.
4. **Fine-tuning Stage Planning:** Transferring learning from the pre-trained model on small labelled datasets to label the pseudo labels produced from pre-trained model.
5. **Evaluation and Optimisation of the model:** Evaluating and comparing model performance with existing methods using performance matrix. Optimising the model with adjustment of calculation function to achieve better result.

3.3 Backbone Architecture Selection

Based on the observation from the literature review in the previous sections, a Convolutional Neural Network (CNN) is the most suitable backbone architecture for this model. CNNs exhibit high efficiency in capturing hierarchical spatial and temporal features from video input, learning from single edges to complex patterns in a layer-by-layer manner [18]. Moreover, CNNs are also capable of learning powerful features from video frames effectively [17], which makes them ideal for facial micro-expressions recognition tasks.

Furthermore, this architecture represents a practical option due to the scarcity of data. Other architectures like Vision Transformers (ViT) are powerful, but there is a high cost in terms of the dataset demanding since it only proves effective when massive datasets are available [3]. Whilst 3D CNNs (like R(2+1)D) are capable of capturing spatial and temporal dynamic features but it come with high computational costs and memory demands [36].

Therefore, a standard 2D CNN backbone provides the optimal trade-off between feature extraction efficiency and computational cost, which is an ideal backbone architecture for training the model on limited unlabelled data and labelled data [40].

3.4 Datasets

Both pre-training and fine-tuning stages require a micro-expressions video dataset. Pre-training typically utilises a large-scale unlabelled dataset for self-supervised learning, while fine-tuning utilises a small labelled dataset[52]. Due to the scarcity of labelled data, most of the micro-expression recognition approaches are benchmarked using standard labelled datasets, including CASME II, SAMM, and SMIC-HS [9, 28, 40, 42]. Each dataset possesses unique characteristics:

1. **CASME II:** A dataset contains 247 validated micro-expression samples where the videos were recorded at a high frame rate of 200 frames per second (fps). The data was collected from 26 valid participants [45].
2. **SAMM:** A dataset contains 159 long videos with different kinds of micro-expressions and macro-expressions, where the videos were recorded at a high frame rate of 200 fps. The data was collected from 32 valid participants [7].
3. **SMIC-HS:** A dataset contains 164 spontaneous micro-expression video samples, captured at a high frame rate of 100 fps. The data was collected from 16 participants [23].

In the fine-training stage of this methodology, the CASME II dataset is utilised as mentioned at section 4.2: Dataset Explanation and Comparison. Each sample in CASME II is annotated with onset, apex, and offset frames, which supports the requirements of the fine-tuning model. Additionally, each sample includes an emotional label with its corresponding activated Action Units (AUs) as defined by the Facial Action Coding System (FACS), making CASME II suitable for the fine-tuning stage. Furthermore, CASME II offers a large number of high-quality videos and comprehensive annotations, which benefit the fine-tuning stages, enabling the model to classify specific micro-expressions from those subtle spatio-temporal features learned from pre-training.

However, the limited number of samples in CASME II often leads to overfitting [40], as self-supervised learning models generally require large datasets to effectively learn spatio-temporal features. To address this limitation, CAS(ME)^3 is employed during the pre-training stage to reduce the risk of overfitting. It is a labelled video dataset that comprises 1109 micro-expressions [21], but this study will use this dataset for the self-supervised pre-training stage by ignoring the labels. Therefore, this methodology will adopt a two-dataset approach: CAS(ME)^3 for self-supervised pre-training stages and CASME II for the supervised fine-tuning stage.

3.5 Pre-training Stage on Unlabelled Video (Upstream Tasks)

Pre-training stage serves as the foundation of the self-supervised learning framework, extracting robust features from large-scale unlabelled facial video datasets without manual annotation [52]. This methodology employs a two-stage pre-training strategy where the feature learned in stage 1 are transferred to stage 2 to learn complementary features: spatio-temporal dynamics of facial structure and discriminative patterns that separate different emotions. As illustrated in Figure 2, the pipelines demonstrate how the model hierarchically learns the feature representations.

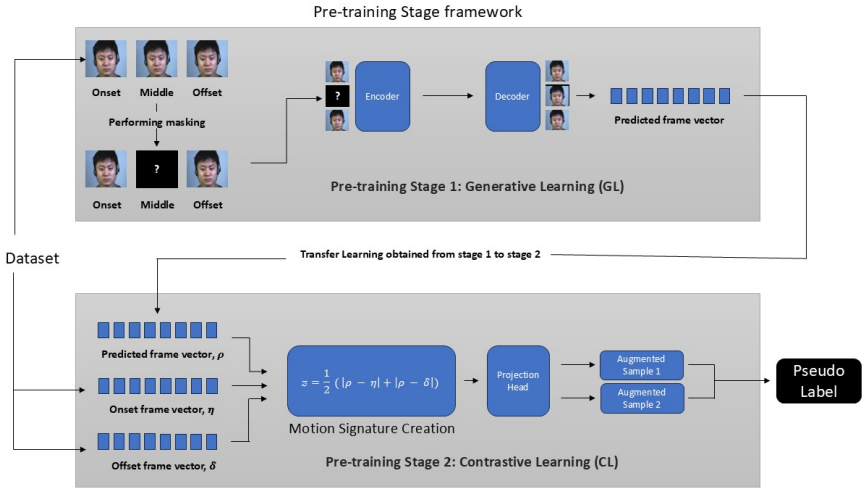


Fig. 2. Pre-training stage

3.5.1 Pre-training Stage 1: Generative Learning (GL) for Motion Spotting

The first stage of pre-training uses generative learning (GL) to train a CNN backbone capable of learning the static structure of the human face and detecting subtle facial motion, which is crucial for identifying the point of maximum intensity in micro-expressions. The objective of the current stage is to enable the model to learn a spatio-temporal representation of facial motion by reconstructing the frames at peak expression intensity (apex frame). Based on the research [24], the apex frame can be detected as the centre of the interval with maximum frequency amplitude. Therefore, we approximate the apex frame by reconstructing the middle frame given the onset and offset frames as input.

The current stage for this methodology is inspired by the motion learning stage of frameworks like SelfME [9] and the powerful spatio-temporal feature learning in VideoMAE [28]. GL has been proven highly successful and particularly beneficial for video-based self-supervised learning [6, 9, 30], motivating this methodology to adopt a generative learning approach. While VideoMAE uses spatial masking to reconstruct random patches

of the video [28], this methodology adopts a temporal reconstruction task, where the model learns to predict middle frames from onset and offset frames.

The architecture consists of a convolutional encoder-decoder structure that was trained on a "temporal in-betweening" task. While the encoder takes the onset frame (X_{onset}) and offset frame (X_{offset}) as input, encoding them into convolutional layers to extract hierarchical features, converting them into 1D feature vectors and the decoder then reconstructs the predicted middle frame ($\hat{X}_{\text{mid}}$). The learning objective for the current stage is to minimise the pixel-wise reconstruction loss between the predicted middle and the ground-truth middle frame. Furthermore, the model is also optimised using a composite loss function that integrates a perceptual loss, forcing generated frame to be realistic and structurally correct.

The composite loss is defined as the function shown below,

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{pixel}} + \lambda \cdot \mathcal{L}_{\text{perc}}$$

Definition of Components:

- $\mathcal{L}_{\textbf{gen}}$ **(Total Generative Loss):** This combined loss function is employed to train the model during the Generative Learning stage.
- $\mathcal{L}_{\textbf{pixel}}$ **(Pixel-wise Loss):** This reconstruction loss quantifies the average absolute difference (L1 loss) between the predicted middle frame ($\hat{X}_{\text{mid}}$) and the ground-truth middle frame (X_{mid}) for each pixel.
- $\mathcal{L}_{\textbf{perc}}$ **(Perceptual Loss):** This reconstruction loss evaluates the semantic and structural differences between the predicted middle frame ($\hat{X}_{\text{mid}}$) and the ground-truth middle frame (X_{mid}).
- λ **(Weighting Factor):** This hyperparameter controls the balance between the Perceptual Loss ($\mathcal{L}_{\text{perc}}$) and the Pixel-wise Loss ($\mathcal{L}_{\text{pixel}}$).

3.5.2 Pre-training Stage 2: Contrastive Learning (CL) for discriminative emotion-pair

The second stage of pre-training adopts Contrastive Learning (CL) to teach the model to distinguish between different types of emotion pairs, forming the foundation for emotion discrimination. The objective is to learn a discriminative embedding space that separates similar micro-expression pairs from different ones, forming emotion pair with pseudo-labels.

The success of the CL approach has been proven in self-supervised learning [19, 31], where a model is trained to pull "positive pairs" (similar samples) closer together while pushing "negative pairs" (dissimilar samples) further apart. This builds upon the generative first stage, as Contrastive learning constitutes a powerful discriminative approach [31].

This stage uses the robust spatio-temporal features learned from stage 1 to continue with emotion-pairing tasks, as shown in the step below:

1. **Feature Extraction:** The pre-trained generative learning model from stage 1 serves as a feature extractor to obtain the representation of the predicted middle frame, which is mostly close to the ground-truth middle frame.
2. **Motion Signature Creation:** The motion signature (z) is calculated by calculating the average of the feature differences between the predicted middle frame and the input frames ($\hat{f}_{\text{mid}}, f_{onset}, f_{offset}$) as shown below:

$$z = \frac{1}{2}\left(\left|\hat{f}_{\text{mid}} - f_{\text{onset}}\right| + \left|\hat{f}_{\text{mid}} - f_{\text{offset}}\right|\right)$$

3. **Contrastive Learning Tasks:** These motion signatures (z) are utilised in contrastive learning tasks to construct positive and negative pairs for different facial video, creating pseudo-labels derived from the data structure.

With the current step, the model is forced to learn to create a clustered feature space, which aligns with the primary goal of contrastive learning — discriminative representations from unlabelled data to improve downstream tasks [19, 41]. Furthermore, the model will be using a contrastive loss function like InfoNCE to optimise the performance [19]:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\exp(\text{sim}(z_i, z_i^+)/\tau) + \sum_{j=1}^{K}\exp(\text{sim}(z_i, z_j^-)/\tau)}$$

Definition of Components:

- z_i **(Anchor):** The feature vector for the current sample i.
- z_i^+ **(Positive Sample):** The feature vector representing a sample that should be "similar" to z_i.
- z_j^- **(Negative Samples)** The feature vector representing a sample that should be "dissimilar" to z_i.
- $\textbf{sim}(u, v)$ **(Cosine similarity):** A function that measures the similarity between two vectors (e.g., ranging from -1 to 1).
- τ **(Tau, Temperature):** A hyperparameter controls the distribution of similarities, regulating how strongly model distinguishes between similar and dissimilar pairs.

3.6 Fine-tuning Stage on labelled Video (Downstream Tasks)

Following the self-supervised pre-training stage, the model proceeds to the fine-tuning stage, utilising the robust and discriminative facial motion representations learned in the previous stage. During fine-tuning, the pre-trained feature extractor is adapted to the downstream task of micro-expression recognition using a labelled video dataset [40, 52]. The

fine-tuning after pre-training approach, also known as Transfer Learning, is a standard paradigm also widely recognised for its effectiveness in adapting generalised, self-supervised features to specific downstream tasks [34, 47].

This methodology will utilise the pre-trained backbone and train the model end-to-end on a labelled video dataset, such as CASME II. Although Wang and Shang [42] investigates five-class recognition, the most widely adopted benchmark for evaluation is three-class classification (Positive, Negative, and Surprise) [9–11, 25]. Therefore, the classification task of the model is planned to train on categorising videos into three emotion classes: Positive, Negative, and Surprise to maximise the measure performance.

Furthermore, this methodology inspired by Shu et al. [39] will be fully fine-tuned with a low learning rate to allow pre-trained features to be adapted to the specific subtleties of downstream tasks. A standard Cross-Entropy Loss will be used as the optimisation objective for this methodology, as it is highly useful and has been demonstrated in similar Micro-expression Recognition fine-tuning pipelines [42].

Therefore, the fine-tuning stages are employed following:

- **Optimiser:** A low learning rate (1×10^{-4})
- **Loss:** Cross-Entropy for class imbalance
- **Strategy**: Fully fine-tuning the entire network
- **Evaluation**: 3-class classification (Positive/Negative/Surprise)

3.7 Evaluations and Optimisation of model

The optimisation of the model is divided into three phases: Generative Learning Optimisation, Contrastive Learning Optimisation and Fine-tuning Evaluation. Each of them employs distinct performance metrics or loss functions to ensure comprehensive model performance across the entire pipeline. Loss function is designed for guiding the model optimisation, while the performance metric is used for measuring model effectiveness. The optimisation method and performance metric used for this study are commonly used by the research community.

3.7.1 Generative Learning (GL) Optimisation

In the pre-training stage 1, generative learning is used for spotting the micro-expressions by reconstructing the middle frame from the onset and offset frames.

- The primary optimisation objectives combine pixel-wise and perceptual losses as defined in section 3.5.1:

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{pixel}} + \lambda \cdot \mathcal{L}_{\text{perc}}$$

  where $\mathcal{L}_{\text{pixel}}$ is measured using L1-norm [9] (Mean Absolute Error) and $\mathcal{L}_{\text{perc}}$ is computed using VGG-based perceptual loss (shown in the research [15]).

- To evaluate the quality of the reconstructed middle frame, the Structural Similarity Index measure (SSIM) (shown in the research [33] is applied.

3.7.2 Contrastive Learning (CL) Optimisation

Contrastive learning in pre-training stage 2 is used for learning the discriminative motion signature of the emotion pair.

- The primary optimisation objectives employs a InfoNCE Loss ($\mathcal{L}_{\text{InfoNCE}}$) [19] as defined in section 3.5.2:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\exp(\text{sim}(z_i, z_i^+)/\tau) + \sum_{j=1}^{K}\exp(\text{sim}(z_i, z_j^-)/\tau)}$$

  where $\text{sim}(u, v)$ computes cosine similarity between vectors and τ is the temperature parameter that scales the similarity distribution
- To evaluate the quality of classification, the Linear Probing Accuracy algorithm is applied [38].

3.7.3 Fine-tuning Evaluation

In the final stage, fine-tuning is applied to micro-expression recognition using labelled data.

- The primary optimisation objectives employ standard multi-class Cross-Entropy Loss ($\mathcal{L}_{\text{CCE}}$):

$$\mathcal{L}_{\text{CCE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c})$$

  where N is the number of samples, C is the number of classes, $y_{i,c}$ is the binary indicator (0 or 1).
- To evaluate the fine-tuning performance, F1-score and Accuracy are measured [10].

4 Feasibility

This section highlights the feasibility evaluation for this study, including the usability of the tools, dataset explanation and comparison, and the summary of the early testing with a small-scale model.

4.1 Evaluation of Tools

This study will implement the model using Python, a widely used programming language with open-source machine learning libraries. Additionally, the primary library of this study will be using PyTorch due to its capabilities and its widespread adoption in the research community, offering tools that support the feasibility of our study. Existing research, such as SelfME [9] and Micron-BERT [34], has shown effectiveness using PyTorch. These results further highlight PyTorch's practicality for robust and accessible deployment in this study.

The Dlib toolkit will be utilised in the data preprocessing pipeline. It automatically identifies and locates 68 specific, pre-defined points on the face and wraps and aligns the face to a standard position [37]. As a standard library for detecting facial key-points in raw video, Dlib enables more accurate feature extraction. Preprocessing is essential for the model, as it aligns and crops raw video to ensure data consistency across all datasets.

4.2 Dataset Explanation and Comparison

Two datasets commonly used for fine-tuning include SAMM and CASME II. Both datasets provide high-quality micro-expression samples with annotated labelling, including apex, onset, and offset frame annotations, as well as micro-expression emotion pairs (Happiness, Surprise, and more), making them suitable for the fine-tuning stage. Both datasets are recorded at 200 frames per second (fps), while CASME II contains 247 validated samples, and SAMM contains only 159. Based on the comparison, CASME II offered a higher data availability, which can increase the feasibility of this study. Therefore, CASME II is selected as the primary fine-tuning dataset for this study, with SAMM as a potential supplementary dataset for future work if necessary.

4.3 Summary of early testing

Initial testing was conducted to evaluate the code's feasibility for this study. Due to limited time, only a simple pre-processing and pre-training stage 1 is conducted as shown below.

In the pre-processing step, preprocessing is performed using Dlib. The image is first synchronised with the CASME II dataset, then preprocessed using dlib and resized. As illustrated in Figure 3, the pre-processed image is standardised and ready for the next step. For the early testing of the methodology, a small-scale feasibility test was conducted to confirm whether the model can learn to reconstruct a middle frame from an onset and offset frame. The current step was implemented using three Python classes:



Fig. 3. Pre-processing stage

1. **set frame class:** This class processes an input dataset of 10 frames by annotating the first frame as the onset frame, the last frame as the offset frame, and the middle frame as the ground-truth frame.
2. **simple encoder-decoder class:** This encoder-decoder class is designed to reconstruct images from input data.
3. **model testing class:** This class loads both the encoder-decoder model and data, and executes training loops using a standard reconstruction loss function.

As illustrated in the Figure 4, this test successfully demonstrated the viability of the generative "temporal in-betweening" task, showing that the model learned to generate a structurally facial image. Additionally, the reconstruction loss dropped from 0.1906 to 0.0346 over 50 training epochs, demonstrating that the model is successfully learning and identifying patterns to improve its predictive accuracy.

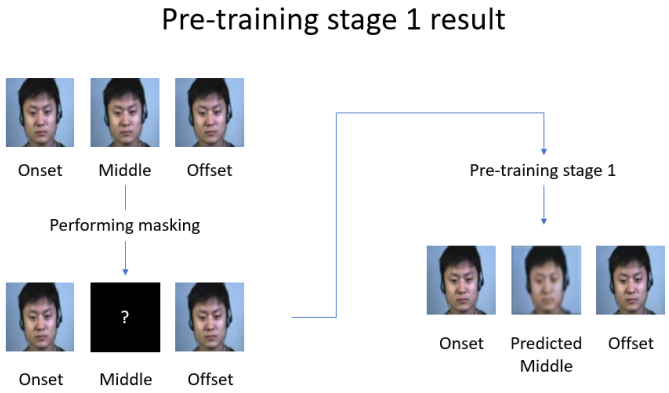


Fig. 4. Generative Learning Testing

4.4 Conclusions

This study proposes a Self-Supervised Learning (SSL) framework to address data scarcity in Micro-expression Recognition (MER). While traditional MER approaches typically depend on large, costly, labelled datasets, this study introduced a two-stage pre-training methodology to extract robust features from unlabelled video data. The proposed framework integrates Generative Learning to capture spatio-temporal features from unlabelled facial video and Contrastive Learning to discriminate emotion pairs.

An initial feasibility test was conducted on a few image samples from the CASME II dataset, demonstrating that Generative Learning can effectively perform the "temporal in-between" task. The model achieved a consistent reduction in reconstruction loss and successfully reconstructed the middle frame given the onset and offset frames. Future work will utilise the CAS(ME)^3 dataset for comprehensive experimentation and full implementation. If time permits, a customised apex frame recognition feature will be developed to enhance model accuracy, rather than relying on the middle frame.

5 Implementation and Contributions

This section details the implementation of the proposed Self-Supervised Learning (SSL) framework for analysing Facial Micro-Expressions (FMEs). It outlines the project pipeline, the chosen architectural design, the data pre-processing and directory-access mechanisms, and the configuration of both the two-stage pre-training and final fine-tuning phases.

5.1 Pipeline Overview

The proposed framework is implemented to extract robust representations from unlabelled video sequences by generating self-supervised signals. By leveraging these self-supervision signals, the model implicitly learns to recognise the underlying dynamics of genuine facial micro-expressions. The comprehensive training pipeline is systematically structured into four distinct stages:

1. **Data Pre-Processing:** Standardising the raw video frames through automated facial landmark detection, noise reduction, and precise geometric centre-cropping to eliminate irrelevant noise.
2. **Pre-Training Stage 1 (Generative Learning):** Training a convolutional encoder-decoder network to predict and reconstruct the middle frame utilising solely the onset and offset frames, forcing the model to learn the underlying spatial facial features.
3. **Pre-Training Stage 2 (Contrastive Learning):** Optimising the pre-trained spatial encoder from stage 1 using a contrastive learning objective. By extracting customised motion signatures of generated middle frame from stage 1, the model learns the complex relational dynamics of subtle facial movements and clusters them into a highly discriminative latent embedding space.
4. **Fine-Tuning Stage (Supervised Transfer Learning):** Transferring the optimised encoder weights to a downstream task. A linear classification head is appended to map the learned feature embeddings into specific emotion categories (Positive, Negative, and Surprise) using the annotated ground-truth datasets.

5.2 Hardware and Software Configuration

The implementation of the model of this self-supervised learning framework were conducted using specific hardware and software environments. To ensure the reproducibility, a complete hardware and software environment are explicitly detailed below.

5.2.1 Hardware Configuration:

While the training of deep Convolutional Neural Networks (CNNs) and the processing of image sequences often utilise Graphics Processing Unit (GPU) acceleration, the experiments for this pipeline were executed entirely within a Central Processing Unit (CPU) environment. The specific hardware setup utilised for this project includes:

- **Operating System:** Windows 11 (64-bit)
- **Graphics Processing Unit (GPU):** NVIDIA GeForce RTX 3050 Ti Laptop GPU (4.00 GB VRAM)

- **Processor (CPU):** AMD Ryzen 5 5600H with Radeon Graphics (3.30 GHz)
- **Memory (RAM):** 16.0 GB

5.2.2 Software and Frameworks

Due to strict dependency constraints and compilation requirements associated with the **Dlib** library, the software architecture was strategically divided into two distinct virtual environments. This separation ensured stability during the complex facial landmark extraction phase and prevented version conflicts during the machine training phase. The specific environments and their corresponding libraries are detailed below:

Data Pre-Processing Environment (Python 3.10):

- **dlib (v19.24) & CMake:** A facial analysis framework compiled using CMake, utilised in conjunction with the shape_predictor_68_face_landmarks.dat model to perform precise 68-point facial landmark detection for facial alignment.
- **OpenCV (opencv-python):** Employed for image input/output operations, grayscale conversion, and the application of bilateral filtering for noise reduction while preserving edge details.
- **NumPy (v1.26.4):** Used for efficient numerical computations, including bounding box coordinate calculations required for geometric centre-cropping.

Pre-training and Fine-tuning Environment (Python 3.11):

- **PyTorch (v2.5.1+cu121) & Torchvision(v0.20.1+cu121):** Served as the foundational deep learning frameworks. They were utilised to construct the 2D-CNN autoencoder, define the contrastive projection heads, and manage the dynamic weight-transfer operations between stages.
- **Scikit-Learn:** Employed to engineer the strict 5-Fold Cross-Validation splits and to compute the rigorous evaluation metrics (Unweighted F1-Score, UAR, and Accuracy).
- **Pandas & NumPy:** Handled the ingestion, parsing, and data manipulation of the complex metadata CSV files corresponding to the CASME II and SAMM datasets.
- **Plotly & Matplotlib:** Utilised to generate both interactive HTML visualisations and static plots for training loss trajectories and cross-validation confusion metrics.

5.3 Data Pre-processing

To ensure the model learns only the robust features of facial movements, a pre-processing script was implemented to eliminate irrelevant background noise. Micro-expression datasets are incredibly sensitive; even a minor shift or removal in pixels can eliminate significant motion data, therefore making a dataset standardisation is mandatory. To standardise the data, strict rules were applied: the nose must be located at the exact centre of the frame, the resolution is fixed at 224x224 pixels, and all vital facial details must be preserved. The automated process is shown as follows:

1. **Bilateral Filtering:** A bilateral filter is applied to every image to reduce background noise while preserving critical edge structures.

2. **Greyscale Conversion:** Images are converted to greyscale to meet the input requirements of the facial detection library (dlib) but will convert back to RGB after.
3. **Facial Landmark Mapping:** The primary subject's face is isolated, and a dlib shape predictor is utilised to extract 68 specific facial landmarks, mapping regions such as the jawline, eyes, nose, and mouth.
4. **Dynamic Center Cropping:** The spatial boundaries of these 68 coordinates are used to calculate the face's centre. A square crop is applied with a 5% padding margin to prevent the loss of peripheral structural features.

As illustrated in Figure 5, the integration of the facial landmark predictor and subsequent centre-cropping mechanism successfully anchored the participant's nose to the exact geometric centre of the frame. This process effectively eliminated extraneous background pixels and enforced strict spatial consistency across the dataset. This standardisation is a mandatory step as convolutional neural networks are highly sensitive to spatial variations and required to capture subtle, high-frequency facial movements characteristic of micro-expressions. The model may learn irrelevant background patterns without this pre-processing step, thereby degrading its ability to extract meaningful motion representations.

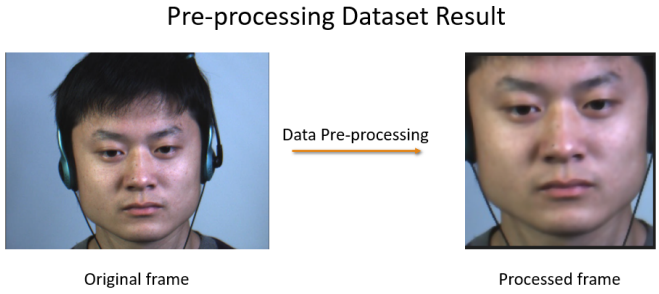


Fig. 5. Result of the pre-processing pipeline, demonstrating successful pre-processed step.

5.4 Robust Directory Access

Because the CASME II, CAS(ME)^2, and SAMM datasets all use different folder structures and file naming rules, a consistent method for loading the data was required. To address this, three separate data-loading scripts were developed to automatically locate and extract the valid micro-expression sequences across the varying formats. Each script includes a custom filtering function that checks the emotion labels and removes any videos that do not belong to the target classes: Positive, Negative, and Surprise. Furthermore, to maintain high data integrity, the pipeline performs an automated frame verification; any sequence identified with missing, corrupt, or inconsistent frame is automatically logged and skipped. This robust alignment mechanism ensures that the temporal relationship between the onset, middle, and offset frames remains synchronised, which is critical for the accurate calculation of motion signatures in subsequent stages.

5.5 Pre-training Stage 1: Self-Supervised Generative Learning for Facial Structure Extraction

The pre-training stage 1 employs a convolutional generative autoencoder to learn the static structural features of the human face. This allows the model to identify subtle facial motions without relying on human-annotated apex frames, predicting the apex as the middle frame of the sequence.

The model performs a "temporal in-betweening" task by processing the onset and offset frames. To capture the temporal dynamics from the outset, the network concatenates the 3-channel onset and offset frames along the channel dimension, creating a single 6-channel input block. This combined input is then processed through a unified encoder. The encoder progressively shrinks the spatial resolution using four convolutional layers equipped with Batch Normalisation and ReLU activations to extract hierarchical spatial features. An Adaptive Average Pooling layer is then applied to spatially compress these feature maps, which are subsequently flattened and passed through a single fully connected layer to generate a unified 512-dimensional latent vector representing the underlying motion dynamics.

A decoder is then implemented to reconstruct a 224×224 pixel approximation of the middle frame from this latent vector. By applying linear expansions and sequential upsampling via transposed convolutions (*ConvTranspose2d*), the spatial dimensions are rebuilt. Finally, a Sigmoid activation function is applied to the output layer to bound the predicted pixel values strictly between 0.0 and 1.0, ensuring format consistency with the normalised input tensors.

The generative model was trained over 100 epochs using the Adam optimiser with a learning rate of 1×10^{-4}. Crucially, the network is optimised by minimising the pixel-wise reconstruction error (L1 Loss) between the predicted middle frame ($\hat{\boldsymbol{y}}$) and the ground-truth middle frame ($\boldsymbol{y}$). This objective is defined mathematically as:

$$\mathcal{L}_{\text{L1}} = \frac{1}{N} \sum_{i=1}^{N} \left| \boldsymbol{y}_i - \hat{\boldsymbol{y}}_i \right| \tag{1}$$

where N represents the total number of pixels in the spatial feature map. By calculating the mean absolute difference between the predicted and target pixels, the L1 loss function heavily penalises structural deviations without excessively blurring high-frequency details. Contrary to the initial proposal, which employed a combination of L1 and SSIM loss functions, the empirical tests indicate that, this architecture only using the L1 loss function alone results in marginally better performance (this finding will be discussed further in Section 6.3.2). Therefore, the primary objective was to minimise this L1 error, with the model successfully converging to a final training loss of approximately 2.25%

.

As illustrated in Figure 6, the visual progression across epochs clearly demonstrates the model's increasing proficiency in generating a middle frame with accurate facial structure positioning. This generative pre-training process ensures that the backbone effectively

extracts robust spatial representations, establishing a strong foundation for the subsequent contrastive learning stage.

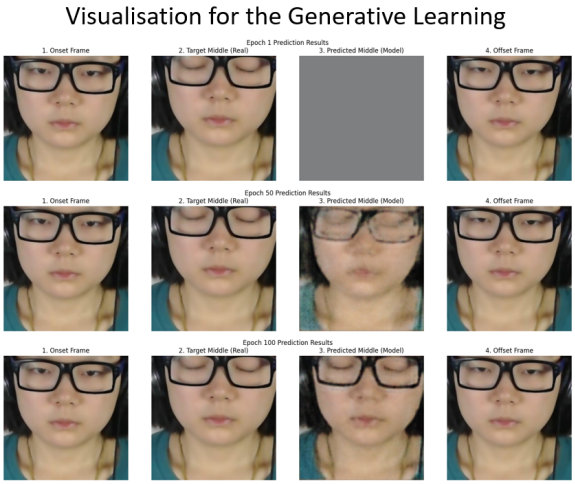


Fig. 6. Visualisation of the Generative Learning progression. The model evolves from predicting a blank latent state (Epoch 1) to capturing rough facial geometry (Epoch 50), and finally reconstructing clear, almost absolute accurate facial features (Epoch 100).

5.6 Pre-training Stage 2: Self-Supervised Contrastive Learning

In this subsequent pre-training phase, the learning paradigm shifts from reconstructing images to learning discriminative representations. By employing a Self-Supervised Contrastive Learning (CL) objective, the model is trained to distinguish the subtle dynamics of varying facial micro-expressions. This contrastive approach naturally groups similar emotional movement patterns together in the embedding space, making it much easier for the final stage to classify emotions without needing massive amounts of manually labelled data.

The model begins by transferring the weights from the pre-trained Stage 1 generative autoencoder to current stage. However, due to the original Stage 1 encoder was designed to accept a concatenated 6-channel input, a structural adaptation is required for Stage 2 to process standard 3-channel RGB motion signatures. By using PyTorch's non-gradient weight copying operations, the pre-trained weights from the first three channels are transferred directly to the newly initialised 3-channel layer while the first convolutional layer is dynamically replaced. It is crucial to acknowledge that this architectural adaptation inherently discards the pre-trained weights by half. Consequently, a portion of the specific temporal dynamics learned during the generative stage is lost which will further be discussed in Section 7.1. However, this trade-off was necessary to standardise the input dimensions while successfully preserving the model's foundational ability to detect low-level spatial edges. Additionally, the Stage 1 decoder is completely frozen and bypassed, as this contrastive task relies exclusively on the encoder for robust feature extraction.

To feed the contrastive learning model, a "motion signature" (z) is mathematically generated. First, the onset and offset frames are passed through the frozen Stage 1 model to predict a middle frame ($\hat{f}_{mid}$). The absolute pixel differences between this prediction and the ground-truth boundary frames are calculated and averaged:

$$z = \frac{1}{2}\left(|\hat{f}_{mid} - f_{onset}| + |\hat{f}_{mid} - f_{offset}|\right) \tag{2}$$

Once generated, this motion signature is subjected to a data augmentation pipeline to create two structurally varied but semantically identical views (Augmentation A and Augmentation B). Specifically, the spatial and pixel-level transformations applied include:

- **Random Horizontal Flipping** ($p = 0.5$) Facial muscle activations frequently exhibit lateral asymmetry. Mirroring the motion signatures forces the network to learn orientation-invariant features, effectively doubling the structural representation of asymmetrical samples
- **Random Resized Cropping** (scaling between 95% and 100% of The dataset include different face size from different participant. A tightly constrained crop prevents the network from memorizing exact pixel coordinates, compelling it to map the relational geometric patterns of motion dynamics instead.
- **Colour Jittering** (Add a 20% of brightness and contrast) It simulate different lighting conditions, increasing the variety of the dataset to simulate different environment of different dataset.
- **Random Rotations** (degree with $-5°$) Subjects rarely maintain perfectly straight head alignment (subject's head is slightly rotated). Applying a minor 5° rotational variance prevents the network from over-optimising for perfectly orthogonal alignments, making the spatial filters robust to natural head tilts.

Figure 7 demonstrates how these combined augmentations alter the motion signature. By applying augmentation, the model is strictly prevented from memorising exact pixel locations, forcing it to learn highly generalisable emotion dynamics.

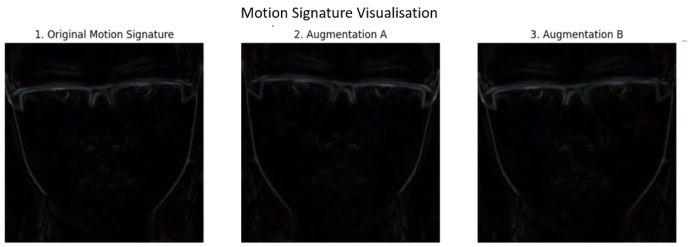


Fig. 7. Visualisation of the Motion Signature Generation and Augmentation. The left image is original motion signature; middle displays Augmentation A; right displays Augmentation B.

These augmented pairs are then passed through the Stage 2 encoder and into a custom projection head. To prevent parameter explosion and over-fitting, this head utilises an

10×10 Adaptive Average Pooling layer, followed by linear transformations, Batch Normalisation, and Dropout regularisation. The specific 10×10 spatial grid was selected based on rigorous empirical testing. As discussed in Section 6.4.2, larger pooling grids drastically increased the complexity and parameter count of the fully connected layers, leading to severe over-fitting where the network simply memorised dataset-specific noise rather than learning generalisable motion patterns. Conversely, smaller spatial grids resulted in under-fitting, as the model lost the fine-grained spatial localisation necessary to detect subtle micro-expressions. This carefully tuned configuration safely compresses the high-dimensional spatial features into a compact 128-dimensional embedding.

The contrastive model was trained over 100 epochs using the Adam optimiser with a learning rate of 1×10^{-4} and a batch size of 8 ($N = 8$). To optimise the network, the InfoNCE loss function is applied. For a single positive pair (i, j) within the augmented batch, the loss is mathematically defined as:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \tag{3}$$

where $\text{sim}(z_i, z_j)$ denotes the cosine similarity, τ is the temperature scalar, and $2N$ represents the total number of augmented views in the batch (16 views). The total objective function, implemented via Cross-Entropy, computes the average loss across all positive pairs:

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^{2N} \ell_{i,p(i)} \tag{4}$$

This objective mathematically pulls the 128-dimensional embeddings of the positive pairs (Augmentation A and Augmentation B of the same motion signature) closer together, while pushing the embeddings of different motion signatures in the batch apart. The model successfully converged to a final stable training loss of approximately 0.86% showing in figure 8, proving it had successfully learned to cluster distinct facial movements.

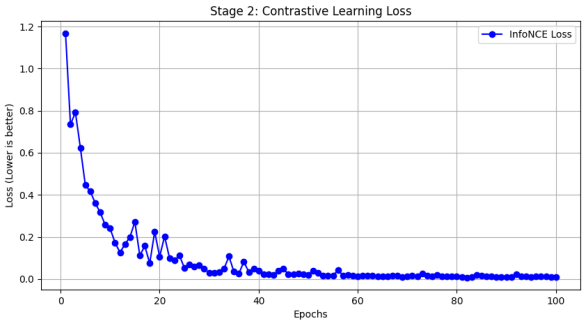


Fig. 8. The loss curve revealed that the model is learning to cluster distinct facial movements

To empirically validate the efficiency of the representation learning, a batch-wise cosine similarity matrix was conducted during the training loop. This matrix plots the relational dynamics of a single forward pass training batch, which comprises 8 unique motion signatures expanded into 16 distinct augmented views. For any two feature embeddings z_i and z_j within this batch, the cosine similarity is mathematically defined as:

$$\text{sim}(z_i, z_j) = \frac{z_i \cdot z_j}{\|z_i\|_2 \|z_j\|_2} \tag{5}$$

As illustrated in Figure 9, the visual progression of this specific batch from Epoch 1 to Epoch 100 reveals a great structural evolution. In these heat-maps, regions of higher cosine similarity are represented by brighter (yellow) intensities. While the matrix at Epoch 1 appears all yellow, but the matrix at Epoch 100 exhibits a highly defined and structured pattern. The primary solid diagonal line in the middle represents perfect self-similarity (where each sample is perfectly correlated with itself). Crucially, because the batch size is 8, the two distinct off-diagonal lines appearing exactly 8 steps away from the centre represent the high similarity of the positive augmented pairs. The dark (purple) regions indicate low similarity, confirming that differing motion signatures are being correctly pushed apart. This highly organised structure visually proves that the model successfully captures meaningful relational dynamics while completely avoiding feature collapse.

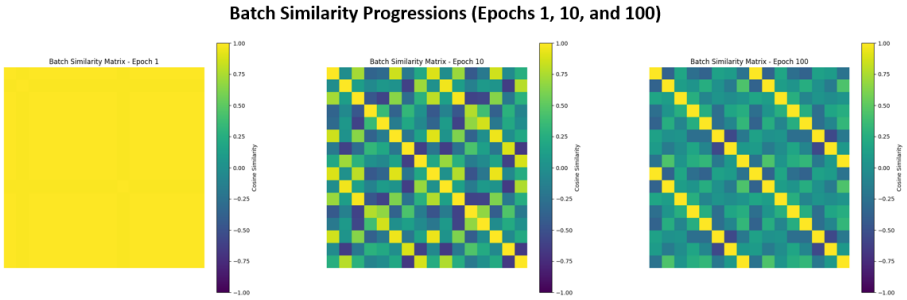


Fig. 9. Progression of the batch-wise Cosine Similarity Matrix. The emergence of the bright off-diagonal lines at Epoch 100 confirms the model has successfully learned to map positive augmented pairs closely together in the latent space.

5.7 Stage 3: Supervised Fine-Tuning for Emotion Classification

To transition the representations learned during the self-supervised pre-training phases to the downstream task, a final supervised fine-tuning stage is implemented. This phase leverages the established components from the earlier stages like the frozen generative autoencoder for middle frame prediction and the contrastive encoder for motion signature extraction to classify genuine micro-expressions into three emotion categories: Positive, Negative, and Surprise.

In this stage, a custom classification head is appended to the pre-trained Stage 2 contrastive backbone. This head consists of a single, fully connected linear layer designed to map the compact 128-dimensional embeddings directly to the three target classes. An end-to-end supervised training approach is executed. To preserve the learned spatial geometry, the Stage 1 generative model remains strictly frozen in evaluation mode. Meanwhile, the Stage 2 spatial encoder and the new linear classifier are jointly trained using a differential learning rate strategy to balance adaptability and stability.

To prevent catastrophic forgetting of the robust contrastive features, the Stage 2 backbone is updated with a highly constrained learning rate of 1×10^{-5}. Conversely, the newly initialised linear classifier is trained with a higher learning rate of 1×10^{-3}, allowing it to rapidly adapt to the latent representations. Both parameter groups are optimised using the Adam optimiser, incorporating a weight decay (L2 regularisation) of 1×10^{-3} to strictly penalise overly complex weights and mitigate overfitting. Furthermore, a Step Learning Rate Scheduler is integrated into the training loop, configured to decay the learning rates by 5% ($\gamma = 0.95$) every 10 epochs. This ensures rapid convergence during the initial epochs while facilitating fine-grained, stable weight updates towards the end of training. This specific hyperparameter configuration was established following extensive empirical tuning, yielding the optimal balance between feature retention and classification accuracy.

To address the severe class imbalance inherent in micro-expression datasets, a weighted Cross-Entropy loss function is employed. The class weights were mathematically derived using the inverse class frequency method (124.0/33.0, 124.0/88.0, 124.0/25.0) to aggressively penalise minority class misclassification. Additionally, the training data is augmented with Random Horizontal Flipping ($p = 0.3$), Random Rotation ($degree = 2$) Random Resized Crop ($scale = (0.97, 1.0)$). A detailed justification for the selection of these specific constrained augmentations is provided in Section 6.6.2. Conversely, the validation dataset is strictly processed without any augmentations, undergoing only necessary resizing to the standard 224x224 resolution. This ensures the evaluation pipeline accurately simulates real-world, unaltered data, providing an unbiased measure of the model's true generalisation capabilities and confirming that it does not merely overfit to the augmented distributions.

The model is trained for a maximum of 60 epochs. Model evaluation is conducted using Accuracy, Unweighted Average Recall (UAR), and the macro-averaged Unweighted F1-score (UF1). The UF1 score serves as the primary selection criterion due to its mathematical robustness in highly imbalanced evaluation settings. To ensure reliable and reproducible generalisation, a strict 5-Fold Cross-Validation strategy is adopted, anchored by a fixed random seed (42). Finally, the best performing model with higher UF1 score will be saved. This mechanism actively monitors the validation UF1 score to halt training and retain the optimally performing model weights.

6 Evaluation and Findings

This section highlights the experimental setup, testing configurations, and performance results of the proposed self-supervised micro-expression recognition model. It provides a comprehensive analysis of the quantitative metrics across all training folds, evaluates the model's predictive behaviour through confusion matrices, and presents a critical discussion on the architectural findings, limitations, and the impact of subject bias on small-scale datasets.

6.1 Dataset Selection

As outlined in the initial project proposal, the framework was originally intended to be pre-trained on the large-scale CAS(ME)^3 dataset and subsequently fine-tuned on CAS(ME)^2. However, due to severe computational resource constraints, processing the extensive CAS(ME)^3 database proved to unfeasible. Furthermore, empirical analysis of the CAS(ME)^2 dataset revealed only 56 valid micro-expression sequences, a sample size significantly insufficient to prevent severe overfitting during supervised training.

Following a critical review of these constraints, the dataset strategy was pivoted. The methodology was restructured to utilise a combination of three widely benchmarked datasets: CAS(ME)^2, CASME II, and SAMM. All datasets were rigorously pre-processed and spatially standardised as detailed in Section 5.3. Under this revised protocol, the self-supervised pre-training stages strictly utilise the micro-expression sequences from CAS(ME)^2, while the supervised fine-tuning and evaluation stages leverage the distinct CASME II and SAMM datasets to test cross-dataset generalisation.

6.1.1 CAS(ME)^2 dataset and Macro-Expression Exclusion

The raw CAS(ME)^2 dataset contains 299 macro-expression sequences alongside with 57 micro-expressions. These macro-expressions were deliberately excluded from this framework because the architecture is explicitly designed to compute and cluster subtle, high-frequency motion signatures, including macro-expressions in the training would mathematically overwhelm the latent space with large-scale facial movements, drastically degrading the model's capacity to learn fine-grained micro-expression dynamics.

Consequently, only the 54 available micro-expression sequences were initially considered for pre-training. Following the automated directory filtering pipeline (detailed in Section 5.4), three sequences belonging to extraneous emotion classes (two labelled as 'Pain' and one as 'Helplessness') were discarded to maintain alignment with the target classification classes. This resulted in a final pre-training on 54 valid CAS(ME)^2 micro-expression sequences.

6.1.2 CASME II and SAMM Refinement

For the supervised fine-tuning and validation phases, the raw CASME II and SAMM datasets initially provided 147 and 125 micro-expression sequences respectively. However, due to the same automated filtering pipeline to eliminate sequences with corrupted, missing,

or unreadable onset/apex/offset frames, one invalid sequence was systematically discarded from each two dataset. This process resulted a final amount of 146 valid sequences for CASME II and 124 valid sequences for SAMM that being used in fine-tuning stage.

6.2 Experimental Setup and Evaluation Metrics

6.2.1 Performance Metrics Selection

As established in Section 5.7, five primary performance metrics were monitored during model evaluation: Accuracy, Unweighted F1-Score (UF1), Unweighted Average Recall (UAR), Training Loss, and Validation Loss. Due to the inherent class imbalance, standard accuracy serves as an insufficient and potentially misleading metric. Therefore, UF1-score will be serve as primary metric for the whole pipeline.

The UF1 score computes the F1-score independently for each class and subsequently averages them, ensuring equal contribution from all classes, including minority classes. This makes it a more reliable indicator of the model's discriminative capability. Model selection was primarily based on higher UF1 scores, supported by stable training and validation loss trends to ensure proper generalisation and to mitigate over-fitting.

6.2.2 Computational Configuration

To accommodate computational constraints, a conservative batch size of 8 was maintained throughout both the pre-training and fine-tuning stages. To ensure strict reproducibility and a consistent evaluation environment across all fine-tuning configurations, the random seed governing the data shuffling process for the 5-Fold Cross-Validation was fixed at 42. Finally, all input frames were strictly standardised to a spatial resolution of 224×224 pixels to maintain dimensional consistency across the network architecture.

6.3 Pre-training Stage 1 Findings

In the stage 1 pretraining, there is two findings that meaningful for the experiment, which is architectural depth and the loss function choice.

6.3.1 Architectural Depth

To determine the optimal representational capacity for the generative autoencoder, an empirical ablation study was conducted to evaluate 3-layer, 4-layer, and 5-layer architectures. Initial testing revealed that a shallow 3-layer network suffered from severe underfitting; it lacked the parameter depth required to reconstruct high-frequency facial details, resulting in elevated training loss and blurred spatial outputs.

Conversely, upgrading to a 4-layer configuration resolved this representational bottleneck. The network successfully extracted the necessary hierarchical spatial features, converging to a highly stable L1 training loss of approximately 2.5% within 100 epochs. While a 5-layer architecture was also considered, it was definitively rejected due to the

principle of diminishing returns. Because the 4-layer model already achieved near-optimal reconstruction fidelity, adding a fifth layer yielded no meaningful performance increase. Instead, it introduced severe computational overhead that threatened to exceed the hardware constraints and increasing the computational time. Thus, the 4-layer model was selected as the optimal balance between computational efficiency and representational power.

6.3.2 Objective Function Optimisation: L1 vs. Composite Loss

As outlined in the initial methodology, the generative autoencoder was originally intended to be optimised using a composite loss function, specifically a weighted combination of Mean Absolute Error (L1 Loss) and the Structural Similarity Index Measure (SSIM). Theoretically, this advanced formulation is highly robust, the L1 component penalises absolute pixel-wise deviations, while the SSIM component enforces global perceptual quality across the frame reconstruction. However, a testing revealed that this composite objective significantly underperformed when applied to the highly specific task of micro-expression reconstruction. As illustrated in Figure 14, the network trained with the composite loss function (left) produced reconstructions that were noticeably degraded, often exhibiting blurred boundaries and failing to capture fine-grained facial topography.

This visual degradation directly contradicted the primary objective of the generative pre-training phase which is reconstructing high-resolution, robust spatial features. The composite loss function inadvertently over-prioritised global perceptual smoothness at the expense of pixel-level edge preservation, resulting a bad pixel-to-pixel reconstructions. Consequently, the composite objective was definitively abandoned in favour of a strict L1 loss function. By penalising absolute error, the L1 objective forced the network to achieve exact geometric alignment. The L1-loss approach drastically improved the sharpness and spatial feature of the generated frames, ensuring the extracted spatial features were optimally pre-processed for the subsequent contrastive learning stage.

6.4 Pre-training Stage 2 Findings

6.4.1 Stage 2 Projection Head: Architecture

The projection head serves as the critical mathematical bridge during the Stage 2 contrastive learning phase, translating high-dimensional spatial feature maps into a compact, discriminative latent space. As previously introduced in Section 5.6, the spatial features extracted by the convolutional generative backbone (comprising 256 channels) are first passed through an Adaptive Average Pooling layer configured to an 10×10 spatial resolution.

Upon passing through this pooling layer, the feature maps are flattened into a continuous 16,384-dimensional vector ($256 \times 10 \times 10$). To aggressively distil this massive feature space into the final representation, the 16,384-dimensional vector is passed through a fully connected linear layer that drastically compresses it to 1,024 dimensions. To ensure gradient stability and prevent overfitting, this layer is immediately followed by Batch Normalisation, a non-linear ReLU activation, and Dropout regularisation ($p = 0.5$). Finally, a second linear

projection compresses the 1,024-dimensional vector down to the ultimate 128-dimensional contrastive embedding used for the InfoNCE loss calculation.

6.4.2 Justification for the 10×10 Spatial Resolution

The explicit selection of an 10×10 Adaptive Average Pooling grid was a architectural choice driven by the need to balance spatial awareness with generalisation capability. In micro-expression recognition, the exact spatial location of a movement is vital for accurate classification. If the pooling layer is too small, the network suffers from severe spatial degradation, permanently losing the fine-grained geometric coordinates of the facial muscles. To empirically validate this, an testing was conducted using a highly compressed 4×4 spatial grid. The results confirmed severe underfitting; the network lacked the spatial resolution required to effectively localise subtle micro-expressions. Consequently, the final Unweighted F1 (UF1) score for the 4×4 model was 15% lower than that of the proposed 10×10 architecture (shown in Figure 15).

Conversely, rigorous empirical testing (shown in Figure 16) revealed that utilising a larger spatial grid, such as 15×15, generated a severely bloated parameter count entering the linear classification layers—escalating from 25600 ($256 \times 10 \times 10$) to 57,600 ($256 \times 15 \times 15$) parameters. On a small-scale dataset like CASME II, this excessive capacity reliably induced severe subject-identity overfitting. Rather than learning universal, generalisable micro-expression features, the network utilised this massive parameter space to memorise the highly specific resting facial topography of the subjects within the training folds. Although the evaluation results indicate that scaling the model from an 10×10 configuration to a 15×15 configuration yields a 2% improvement in performance, but the significant divergence between the training loss and validation loss trajectories suggests that larger models are prone to overfitting (shown in Figure 16. This phenomenon suggests that the 15×15 configuration likely captures noise specific to the dataset rather than learning generalisable representations.

6.5 Motion Signature Alignment

During the optimisation of the pipeline, a critical empirical adjustment was made to the spatial boundaries of the generated motion signatures. Initially, the signature was calculated by averaging the absolute pixel differences between the predicted apex (middle) frame and the temporal boundary frames (onset and offset). However, visual inspection revealed that these raw motion signatures captured significant task-irrelevant noise around the facial boundaries and neck, as shown in Figure 7. To resolve this, a custom spatial cropping function was introduced to discard 10% of the marginal pixels from the left, right, and bottom edges of the spatial tensor.

This targeted crop effectively eliminates extraneous background noise, compelling the network to focus exclusively on the core facial regions where genuine micro-expressions occur. As illustrated in appendix 13, the adjusted motion signatures demonstrate superior spatial alignment, cleanly isolating the semantically relevant facial dynamics. By

preventing the model from learning spurious correlations tied to background artifacts or macro-movements in the neck, this alignment strategy led to an immediate performance improvement, increasing the average UF1 score to 68%. Therefore, this motion signature adjustment was applied across the Stage 2 contrastive pre-training.

6.6 Pre-training Stage 3 Findings

6.6.1 Stage 3 Supervised Fine-Tuning Findings: Classifier Complexity

To transition the model from self-supervised representation learning to the downstream supervised task, a classification head was required to map the 128-dimensional latent vector to the three target emotion classes (Positive, Negative, and Surprise). To empirically determine the optimal classifier configuration, a test was conducted by comparing a standard single linear layer against a deeper 3-layer Multi-Layer Perceptron (MLP). This experimental MLP head was designed to progressively compress the latent representations through sequential hidden layers ($128 \rightarrow 64 \rightarrow 32 \rightarrow 3$). However, the result of the training dynamics shown in figure 12 revealed that this deeper classification head was destroying the model stability. The training and validation loss trajectories became highly unstable, inducing an immediate and severe overfitting problem.

This performance collapse is directly attributable to severe over-parameterisation. As detailed in previous section, the Stage 2 contrastive projection head had already subjected the spatial features to a massive parameter transformation (compressing a 25600-dimensional (256 * 10 * 10) flattened grid down to 1,024, and finally to 128). Appending an additional 3-layer MLP to classify a micro-scale dataset of only 247 valid sequences provided the network with too much of trainable parameters. Rather than learning to strictly classify the robust motion signatures extracted during pre-training, the MLP utilised this excess capacity to memorise the highly specific, subject-level features of the training folds. This aggressive memorisation completely destroyed the model's capacity to generalise to unseen subjects during the validation. Therefore, the deeper MLP architecture was abandoned while a single linear classifier is remained for the framework.

6.6.2 Constrained Data Augmentation for Motion Signatures

To increase the variety of the dataset, a constrained augmentation pipeline was designed to safely expand the structural variance of the motion signature without compromising the real-world fidelity of the extracted motion signatures. The specific transformations applied include:

- **Random Horizontal Flipping (p=0.3):** Facial muscle activations frequently exhibit lateral asymmetry. Mirroring the motion signatures forces the network to learn orientation-invariant features, effectively doubling the structural representation of asymmetrical samples.
- **Random Resized Cropping (Scale: 97% to 100%):** The dataset sometimes alter the absolute bounding box of the face. A tightly constrained crop prevents the

network from memorizing exact pixel coordinates, compelling it to map the relational geometric patterns of motion dynamics instead.

- **Random Rotations (degree with ±2°):** Subjects rarely maintain perfectly straight head alignment (The patience's head is slightly rotated). Applying a minor 2° rotational variance prevents the network from over-optimising for perfectly orthogonal alignments, making the spatial filters robust to natural head tilts.

These augmentation during the fine-tuning stage must be lighter compared to the Stage 2 model to prevent destroying the subtle features learned from stage 2. More aggressive augmentations will severely degraded performance. For instance, Random Gaussian Blur smoothed out subtle FMEs, Random Grayscale obscured motion intensity, and Random Vertical Flip completely inverted natural facial structures which is not useful for model training. However, even thought the applied augmentations only yielded a modest 5% improvement in the UF1-score, they successfully mitigated early overfitting. Therefore, while data augmentation is not the primary focus of this framework, it serves as a necessary and effective regulariser.

6.7 Final Performance Results

6.7.1 5 Fold Validation Result

5-fold cross-validation provides an efficient mechanism for model evaluation, but finding direct state-of-the-art comparisons for a 3-class Micro-Expression Recognition (MER) task under this specific split is challenging within current literature. Consequently, Leave-One-Subject-Out (LOSO) cross-validation remains the gold standard for final benchmarking in MER, as it rigorously evaluates a model's ability to generalise to entirely unseen subjects, thereby preventing the network from biasing towards subject-specific facial morphologies. However, running a full LOSO evaluation pipeline for every architectural adjustment is computationally prohibitive and practically unfeasible within the time and resource constraints of this study. Therefore, a two-stage validation strategy was adopted.

First, an efficient 5-fold cross-validation ($k = 5$, with the random seed fixed at 42 to ensure reproducibility) was utilised as a rigorous quality-check and hyperparameter tuning environment. This tuning phase was conducted independently on the CASME II and SAMM datasets. Once the optimal model configuration and hyperparameters were identified using the 5-fold CV, the single best-performing architecture was selected to undergo the computationally intensive LOSO validation.

As presented in Table 1, the model achieved an average UF1 score of **67.80%** on the CASME II dataset. However, a notable performance variance was observed across the individual splits, ranging from a low of **55.77%** (Fold 3) to a peak of **76.94%** (Fold 2). This gap indicates that the model's discriminative capability is sensitive to the specific distribution of micro-expressions within a given fold and struggle when a validation split contains inherently more ambiguous or subtle samples. This struggling also happened in the SAMM dataset, which yielded an average UF1 of **62.88%**. While Fold 4 achieved a highly competitive

Table 1. Performance comparison on CASME II and SAMM datasets using 5-fold cross validation.

Fold	CASME II (%)			SAMM (%)		
	UF1	UAR	Acc	UF1	UAR	Acc
Fold 1	64.55	66.67	65.52	73.82	71.30	72.00
Fold 2	76.94	81.40	82.76	62.02	61.27	60.00
Fold 3	55.77	60.00	65.52	67.26	67.62	60.00
Fold 4	71.15	71.83	72.41	75.54	79.17	76.00
Fold 5	70.59	73.28	75.86	35.78	40.56	54.17
Average	**67.80**	**70.64**	**72.41**	**62.88**	**63.98**	**64.43**

75.54%, but Fold 5 exhibited a significant performance drop to **35.78%**. These results validate the decision to use the 5-fold cross-validation merely as a hyperparameter tuning environment; the observed instability underscores the absolute necessity of advancing to a strict Leave-One-Subject-Out (LOSO) protocol to accurately and fairly benchmark the model's true, subject-independent generalisation capabilities.

6.7.2 Leave-one-subject-out (LOSO) Validation Result

To evaluate the performance of the proposed Self-Supervised Learning (SSL) framework, the Leave-One-Subject-Out (LOSO) cross-validation results were benchmarked against five established Micro-Expression Recognition (MER) like handcrafted feature extractors (LBP-TOP[51], Bi-WOOF[27]), a 3D Deep Learning network (DPN+STSTNET[44]), a Graph-based network (AGANTF[48]), and a highly complex SSL-based Transformer model (TGMAE[29]). models across the CASME II and SAMM datasets.

As illustrated in Table 2, the proposed model achieved the lowest UF1 and UAR scores on the CASME II dataset, trailing the top-performing method by approximately 28%. However, on the SAMM dataset, the proposed method outperformed handcrafted baseline methods such as LBP-TOP and Bi-WOOF. Because LBP-TOPạnd Bi-WOOF were largely optimised around the characteristics of earlier datasets, this finding confirms that the proposed self-supervised model possesses genuine cross-dataset generalisation capabilities, even if it has not yet reached its maximum predictive potential. The performance gap between proposed framework and state-of-the-art can be attributed to three architectural constraints:

(1) **Absence of Optical Flow Inputs:** The proposed model relies solely on standard RGB frames. In contrast, highly effective deep learning MER architectures, such as AGANTF and DPN+STSTNET, extracting and utilising optical flow features before feeding the data into a neural network. Optical flow isolates subtle facial dynamics

Table 2. Performance Comparison against State-of-the-Art on CASME II and SAMM Datasets

Model	**Category**	**CASME II**		**SAMM**	
		UF1	**UAR**	**UF1**	**UAR**
LBP-TOP [51]	Handcrafted	0.7026	0.7429	0.3954	0.4102
Bi-WOOF [27]	Handcrafted	0.7805	0.8026	0.5211	0.5139
DPN + STSTNET [44]	Deep Learning (3D)	0.8522	0.8751	0.7724	0.7795
AGANTF [48]	Graph-based	0.9024	0.9221	0.7928	0.7643
TGMAE [29]	SSL-based	**0.9597**	**0.9562**	**0.8415**	**0.7885**
Proposed Model	**SSL-based**	0.6780	0.7064	0.6288	0.6388

far more effectively than raw spatial pixels.

(2) **Temporal Modeling Constraints:** Utilising a 2D CNN as the primary feature extractor restricts the model's capacity to inherently capture sequential temporal dynamics. Top-performing networks leverage 3D CNNs (e.g., DPN+STSTNET) or Spatio-Temporal Transformers (e.g., TGMAE) to natively encode the continuous temporal evolution of micro-expressions. This resulting a big cap between these top-performing networks and the proposed model.

(3) **Pre-training Scale Limitations:** State-of-the-art models benefit heavily from massive datasets; for instance, TGMAE utilizes the large-scale DFME dataset for pre-training, and AGANTF leverages extensive facial graph data. The proposed model was pre-trained exclusively on the relatively small CAS(ME)2 dataset with only micro-expression sequences. This indicates that the scale of the pre-training dataset was insufficient to extract the robust, highly discriminative feature representations required to compete with more complex architectures.

These three architectural constraints are the primary factors limiting the current model's performance, identifying them can provides valuable direction for future research. Therefore, to achieve state-of-the-art performance in future, these foundational design limitations must be addressed by transitioning toward more robust spatio-temporal architectures, integrating optical flow dynamics, and significantly expanding the self-supervised pre-training paradigm.

7 Discussion and Conclusion

This section will highlight the limitations of the proposed framework, outline future work with actionable insights, and provide the final conclusion.

7.1 Limitations

While the proposed self-supervised framework successfully extracted discriminative motion signatures, several inherent limitations must be acknowledged:

- **Pre-training Scale and Persistent Overfitting:** As observed in the benchmark comparisons, the proposed model did not reach state-of-the-art (SOTA) performance. A primary factor is the restricted scale of the pre-training dataset. Self-supervised learning (SSL) heavily relies on massive, diverse datasets to learn robust, generalisable representations. Because the pre-training was limited to the CAS(ME)^2 dataset, the model lacked the data volume necessary to prevent overfitting. Consequently, despite the implementation of aggressive regularisation and early stopping, the network exhibited a tendency to memorise training samples rather than generalising to unseen subjects which is a huge design flaws.

- **Stage-to-Stage Knowledge Transfer and Information Loss:** A significant design constraint within this pipeline is the transition mechanism between Stage 1 and Stage 2. While Stage 1 successfully learns complex facial features to reconstruct the middle frame, but because of the motion signature subtraction needed in stage 2, the stage 1 model is forced to discards a substantial portion of the high-dimensional latent knowledge (the network weights) acquired to compute the motion signature subtraction. This creates an information bottleneck, leading to a computational inefficiency where valuable spatial understanding is lost.

- **Dimensionality Constraints (2D vs. 3D Spatial-Temporal Modelling):** The proposed framework relies exclusively on 2D Convolutional Neural Networks. However, micro-expressions are inherently dynamic, and capturing their crucial temporal dimension is mandatory. The current framework attempts to compress this temporal shift into a single 2D spatial motion signature. This reductive compression results in the model failing to capture the continuous, fine-grained temporal evolution and did not perform well like the advanced architectures, such as 3D CNNs or Spatio-Temporal Transformers. By flattening the temporal domain, the model inherently loses a massive amount of meaningful dynamic features.

- **Algorithmic Simplicity and Scope Limitations:** Due to the scoping and time constraints of this study, the feature extraction techniques and loss formulations employed were relatively foundational. Advanced methodologies highly prevalent in SOTA micro-expression recognition such as dense optical flow extraction, attention mechanisms, or graph-based facial landmark routing; all of these advanced methodologies were not integrated. Relying purely on raw RGB differences of this

framework resulted in an overly simplistic representation of micro-expressions, leaving significant room for methodological enhancement.

7.2 Future Work

To address the mentioned limitations discussed in previous section, several architectural and methodological improvement are proposed for future iterations:

- **Transition to 3D Spatio-Temporal Architectures:** Transition to 3D CNN Architectures: Instead of relying on 2D CNNs and subsequently discarding valuable latent weights from the generative phase (Stage 1), future models should integrate 3D CNNs using the valuable latent weights. This architectural upgrade would allow the network to process temporal feature to compute spatio-temporal motion signature without losing the high-dimensional spatial knowledge acquired during the initial pre-training.

- **Cross-Domain Pre-training on Macro-Expressions:** The current framework underutilised the macro-expression samples available within CASME II dataset. Future iterations should implement a cross-domain training strategy: pre-training the contrastive backbone on massive datasets with macro-expressions to learn robust facial representations. The model can then be fine-tuned on micro-expressions, allowing the model to learned more robust features from large scale of dataset.

- **Integration of Optical Flow Dynamics**: As demonstrated by top-performing state-of-the-art models, optical flow is a highly efficient modality for extracting facial dynamics. Future frameworks should combine the currently proposed motion signatures with dense optical flow estimation between consecutive frames. This multi-modal fusion would explicitly guide the network's attention toward active muscle deformations, significantly reducing the impact of static background noise.

7.3 Conclusion

This study successfully implemented a Self-Supervised Learning (SSL) framework designed to extract and classify subtle Facial Micro-Expressions (FMEs) without relying on heavily annotated apex frames. While the framework encountered specific architectural and computational constraints, it demonstrated good cross-dataset generalisation capabilities. Notably, the model outperformed established handcrafted baseline methods on the SAMM dataset even though underperformed to other SOTA methods. This still indicates that the framework successfully learning to extract robust feature from the dataset.

However, the evaluation also highlighted critical limitations restricting the model from achieving state-of-the-art (SOTA) performance on the CASME II dataset. The framework's reliance on a 2D CNN backbone inherently restricted its capacity to model continuous temporal evolution, forcing a reductive compression of dynamic movements into a single

spatial motion signature. Furthermore, the limited scale of the pre-training dataset constrained the model's representational depth. Addressing these bottlenecks provides clear pathways for future iterations. By transitioning to 3D spatio-temporal architectures, integrating dense optical flow to guide feature attention, and scaling the pre-training domain to include macro-expressions, future models can natively encode the complex, continuous dynamics of human micro-expressions and getting close to SOTA performances.

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A Project Management

A.1 Gantt Chart (Semester 1)

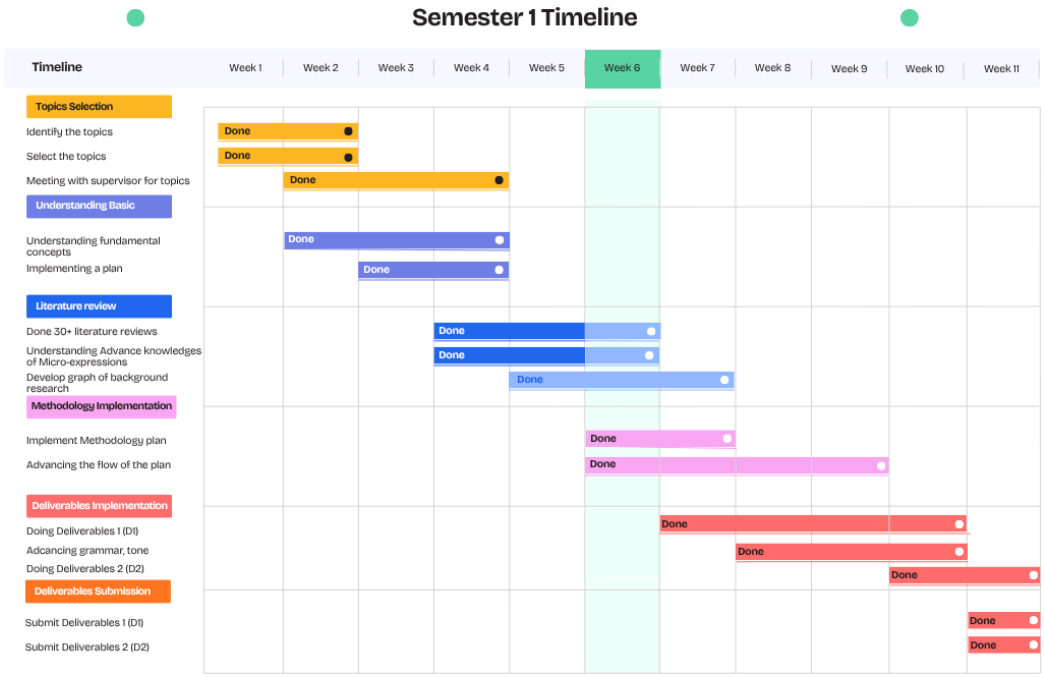


Fig. 10. Gantt Chart for Semester 1

A.2 Gantt Chart (Semester 2)

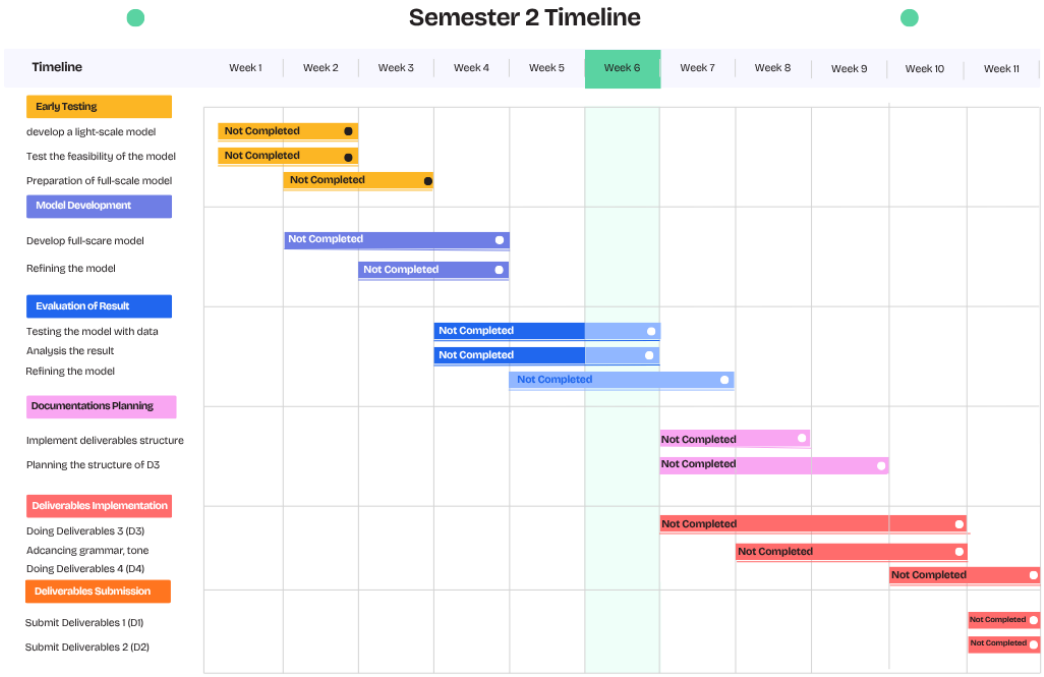


Fig. 11. Gantt Chart for Semester 2

A.3 Risk Analysis

This section highlights the potential risks that may arise during the fine-tuning project. Each identified risk is evaluated based on its likelihood and impact on the project. Three tables are provided for clarity:

The table 3 highlights each category of the risk priority with colour and definition.

Risk Priority	**Colour**	**Category Definition**
Low		The issue has minimal impact to the project, it can be handled with normal monitoring
Medium		The issue has moderate impact to the project, it requires some time to be handled to prevent escalation
High		The issue has a critical impact to the project, immediate actions are compulsory to be taken to handle the issue

Table 3. Risk Priority Category

The table 4 lists each risk along with its probability of occurrence and impact rating.

Risk	**Likelihood**	**Impact**
overfitting during fine tuning due to small dataset	Medium	High
Insufficient labelled data for fine-tuning training	Medium	High
Incorrect feature learning during pre-training stage	Medium	Medium
Incorrect pre-processing step	Medium	Medium
High computational cost during training	Medium	Low
Hardware limitation for model training	Low	Medium
Insufficient time to complete project	Medium	Medium
Potential ethical issues on dataset usage	Low	Low

Table 4. Risk Analysis for the proposed project

The table 5 outlines the risk with its mitigation strategy to handle each issues.

Risk	**Mitigation Strategy**
overfitting during fine tuning	Apply regularisation techniques to model's generalisation capability
Insufficient labelled data	Train the model using CAS(ME)^3 dataset during fine-tuning stage
Incorrect feature learning	Using a smaller dataset for validation before training and perform visual inspection for reconstructed frames
Incorrect pre-processing	Implement a standardised pre-processing pipeline for facial data and validate the output of pre-processing across sample data
High computational cost	Use a lightweight CNN architectures
Hardware limitation	Utilise university lab resources
Insufficient time	Perform early validation and feasibility test before full-scale experiments.
Potential ethical issues	Strictly use only publicly available, ethically approved datasets

Table 5. Risk with mitigation strategy

A.4 PLES Analysis

This sections highlights the analysis of Professional, Legal, Ethical, and Social Issues (PLES) pertinent to the project.

Professional: This project follows established computing standards that require integrity and professionalism to ensure credibility. The dataset has been validated and widely used by the research community, showing its professionalisation. A deep literature review is completed and documented, ensuring the project can be reviewed by the research community.

Legal: All datasets used are fully compliant with regulations. Additionally, any personal data will be anonymised. The project will adhere to the licenses of the open-source frameworks, such as TensorFlow, PyTorch, and more. The project is validated by the university and does not allow for commercial use or other illegal use.

Ethical: The datasets used have been obtained with an ethical certificate. Data must not be misused, misrepresented, or repurposed without appropriate consent. Participants who contributed to the dataset have been informed and accepted to a consent form. This project collects only the necessary data from the dataset for the selected feature.

Social: This project may face the issue of the model being reused for illegal purposes. However, the project aims to deliver good information and make a contribution to the research community. Misuse of the model may raise public concern and have an unexpected impact on society.

B Generative AI

This section outlines the use of Generative AI in the context of this project.

Generative AI (GenAI) is only used for permitted purposes, and each usage is listed as shown below:

- GenAI is used for rephrasing the section to enhance clarity, tone, or grammar.
- GenAI is used for generating ideas and provides critical feedback when misunderstanding flawed technique concepts.
- GenAI is used to assist in code testing by identifying bugs and offering suggestions for improvement.
- GenAI is used for providing suggestions for LaTeX formatting to support the completion of Deliverable 1 (D1) and Deliverable 3 (D3)

No fully AI-generated paragraphs, extensive code implementations, or content with unsupported article are present in this deliverable.

C Project proposal and research changes

This section outlines the summary of any changes from D1 to D3.

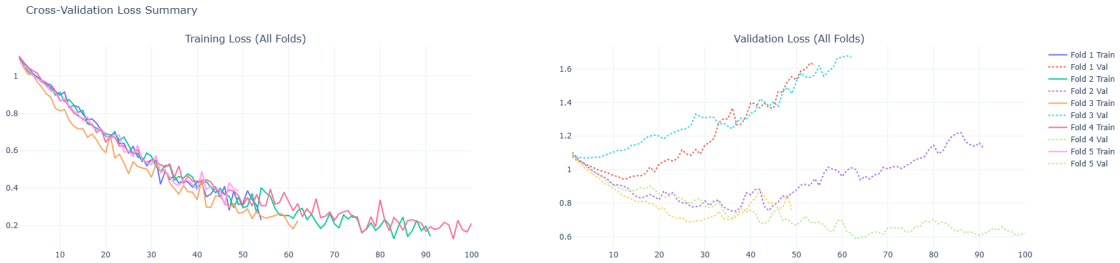
- **Dataset Change:** Originally proposed to utilise the large-scale CAS(ME)^3 dataset for self-supervised pre-training. Due to resource and computational limitation, the framework was revised to use CAS(ME)^2 for pre-training and utilise CASME II and SAMM for downstream fine-tuning and cross-dataset evaluation.
- **Stage 1 Loss Function Adjustment:** Originally proposed a composite loss function for the Stage 1 generative autoencoder. However, empirical testing revealed that utilising L1 loss proven to have better performance compared to composite loss function. Therefore, the objectives was modified to use a L1 loss for training.
- **Stage 3 Loss Function Adjustment:** The proposal D1 planned to use a standard multi-class Cross-Entropy Loss for the fine-tuning stage. This was changed in the final implementation to a weighted Cross-Entropy Loss to aggressively address the severe class imbalance inherent in the datasets.
- **Stage 3 Learning Rate and Fine-Tuning Strategy:** The initial plan was to fully fine-tune the entire network using a single, low learning rate of 1×10^{-4}. However, the final model instead adopted a differential learning rate strategy: the Stage 2 backbone was updated at a highly constrained 1×10^{-5}, while the new classifier used 1×10^{-3}. A Step Learning Rate Scheduler was also introduced to decay rates by 5% every 10 epochs.
- **Motion Signature Alignment (Stage 2):** While the mathematical extraction of the motion signature remained consistent with the proposal [cite: 145, 696], the final implementation introduced a custom spatial cropping function[cite: 334, 337]. This function discards 10% of the marginal pixels from the edges to eliminate task-irrelevant background and neck noise, which successfully improved the average UF1 score[cite: 337, 338, 340].

D Appendix

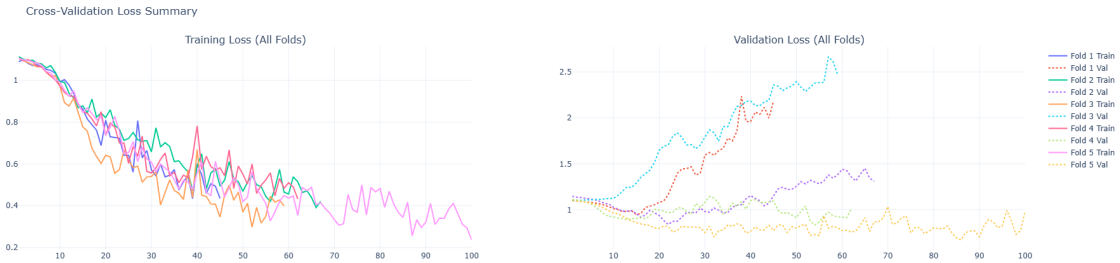
D.1 GitHub Repository

The implementation of the proposed model is publicly available at the following GitHub repository: https://github.com/JeffKian14/FYP-SSLAFM/tree/master/FYP-SSLAFM/Scripts/model_standard

D.2 Figure

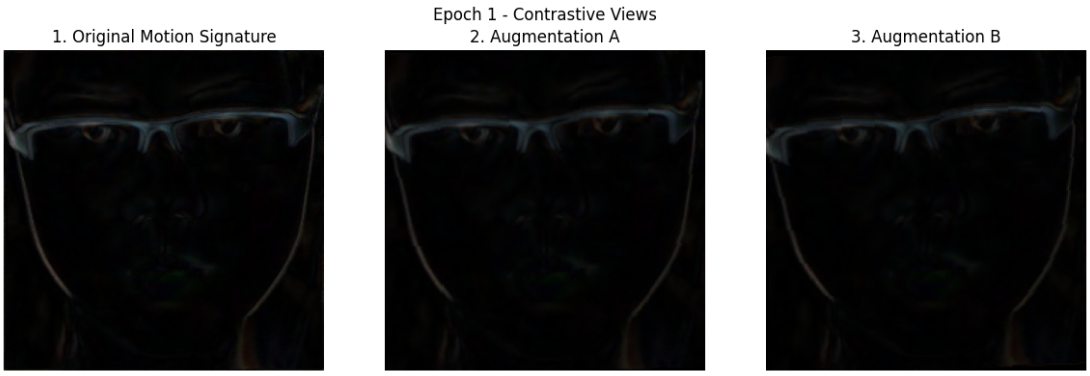


a) Single Linear Classifier: Stable Convergence

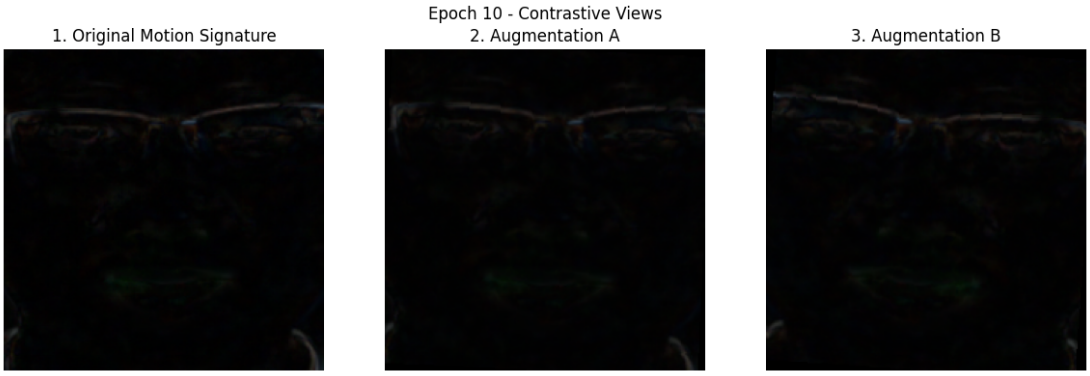


b) 3-Layer MLP Classifier: Unstable Convergence

Fig. 12. Comparative analysis of training and validation loss trajectories during Stage 3 fine-tuning. The Multi-Layer Perceptron (b) exhibits severe gradient instability and identity memorisation, whereas the proposed single linear layer (a) demonstrates smooth, stable convergence, even thought both showing the overfitting problem but MLP is more unstable.

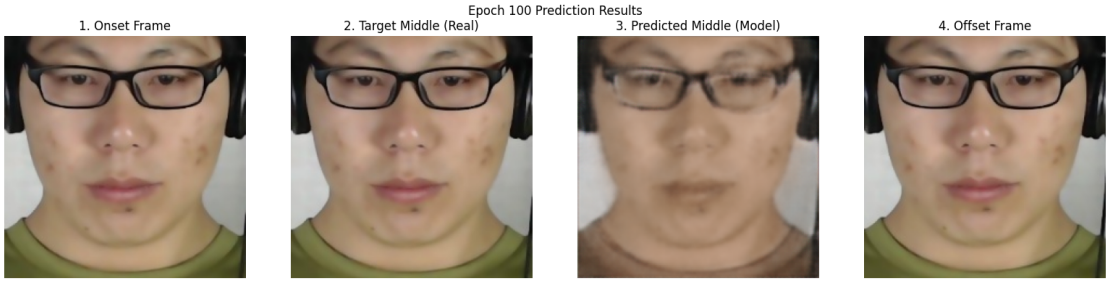


a) Original Motion Signature

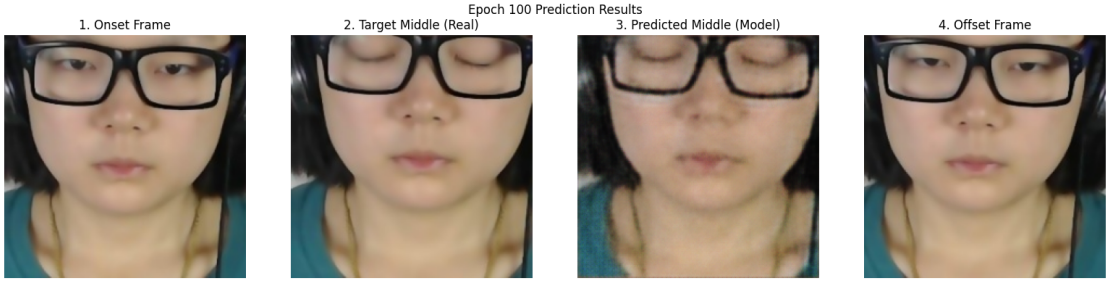


b) Modified Motion Signature

Fig. 13. Comparison between modified motion signature alignment and original motion signature. The original motion signature captured background noise like neck, while the modified motion signature only reveal the facial structure.



a) Composite Loss (L1 + SSIM)

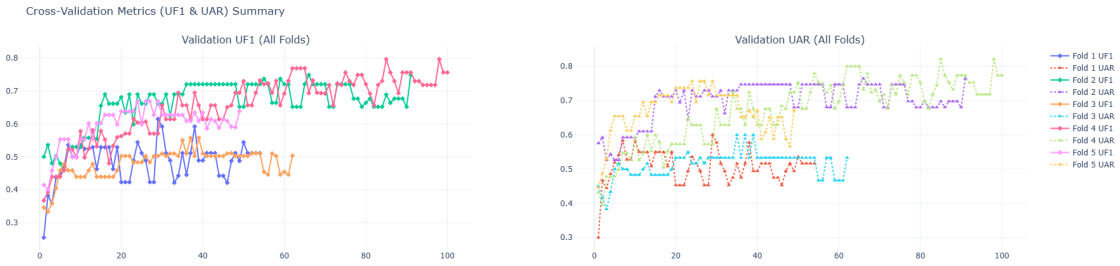


b) L1 Loss

Fig. 14. Visual comparison of middle-frame reconstructions at Epoch 100. The composite loss (a) results in less defined facial boundaries, whereas the L1 loss (b) successfully reconstructed sharp spatial geometry.

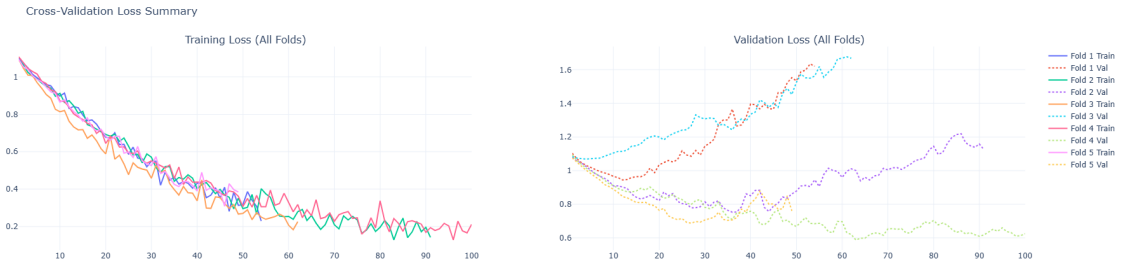


a) 4×4 layers architecture's performance graph

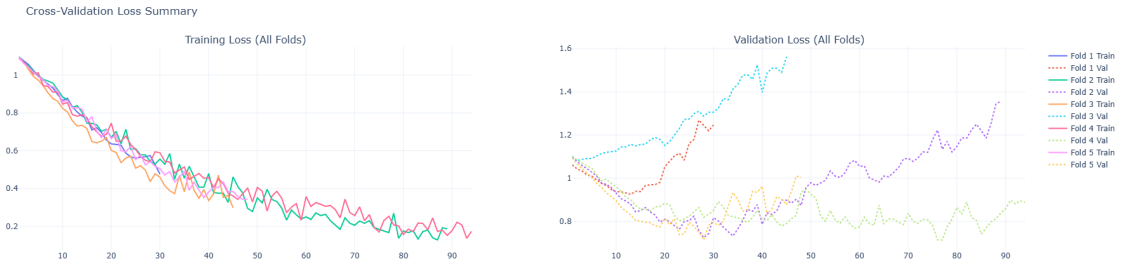


b) 10×10 layers architecture's performance graph

Fig. 15. Comparison of the performance for 5 fold between 4×4 and 10×10 architecture. While 10×10 architecture hitting highest of 79.65%, the 4×4 architecture hitting only 71.89%. Moreover, the average of the validation UF1 of the 4×4 architecture is lower than the 10×10 architecture with 9% different.



a) 10×10 layers architecture's loss graph



b) 15×15 layers architecture's loss graph

Fig. 16. Comparison of the loss for 5 fold between 15×15 and 10×10 architecture. The 15×15 architecture showing a more unstable validation loss and training loss compared to the 10×10 architecture